

## Rectifiers

1. The input of a full-wave rectifier is a cosine voltage with peak  $V_M = 5$  V and frequency 60 Hz, and output load resistance is  $R = 2$  k $\Omega$ . Silicon diodes are used in this circuit for which the forward drop is  $V_{D0} = 0.7$  V.

- (a) Briefly explain the purpose of a rectifier and describe its operation. [1]  
(b) Show the input and output waveforms. [2]  
(c) Calculate the DC value of the output voltage. [1]

Now after connecting a capacitor in parallel with the load, the output becomes a ripple voltage  $V_{out} = V_{DC} \pm 0.2$  V

- (d) Calculate the **peak-to-peak ripple voltage**, and from that, the value of the capacitor. [2]  
(e) Calculate the average of the output voltage  $V_{DC}$  after connecting the capacitor. Compare this with the DC value determined in 'c' and comment on the difference between these two. [2]

2. The input of a **Half-wave rectifier** is a sine voltage with peak  $V_M = 10$  V and frequency 55 Hz, and output load resistance is  $R = 2.5$  k $\Omega$ . Silicon diodes are used in this circuit for which the forward drop is  $V_{D0} = 0.4$  V.

- (a) Calculate the DC value of the output voltage. [1]  
Now after connecting a capacitor in parallel with the load, the output becomes a ripple voltage  $V_{out} = V_{DC} \pm 0.3$  V.  
(b) Calculate the **peak-to-peak ripple voltage**, and from that, the value of the capacitor. [2]  
(c) Draw the **Voltage Transfer Characteristic (VTC) curve** [2]

3. The input of a full-wave rectifier is expressed by,  $V_s(t) = 7\sin(400\pi t)$ , and output load resistance is  $R = 5$  k $\Omega$ . Silicon diodes are used in this circuit for which the forward drop is  $V_{D0} = 0.3$  V.

- (a) Calculate the input and output wave frequency. [2]  
(b) Show the input and output waveforms.

Now after connecting a capacitor,  $C = 100$   $\mu$ F in parallel with the load.

- (c) Calculate the peak-to-peak ripple voltage,  
(d) How can you provide better filtering for the output waves?  
(e) What is the frequency of the Ripple voltage?

4. The input of a **Half-wave rectifier** is a **Square** wave voltage with peak  $V_M = 15$  V and frequency 0.5 Hz, and output load resistance is  $R = 5$  k $\Omega$ . Silicon diodes are used in this circuit for which the forward drop is  $V_{D0} = 0.7$  V.

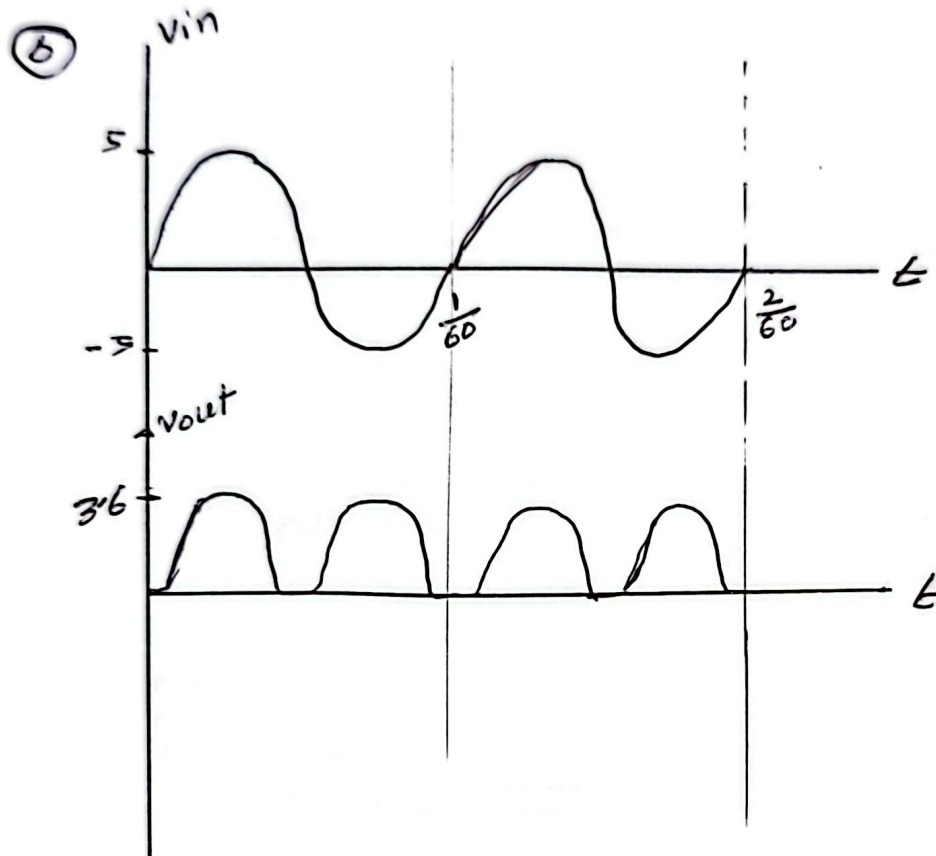
- i. Show the input and output waveforms. [4]  
ii. Draw the VTC curve [2]

5. The input of a **full-wave rectifier** is a **Square** wave voltage with peak  $V_M = 15$  V and frequency 0.5 Hz, and output load resistance is  $R = 5$  k $\Omega$ . Silicon diodes are used in this circuit for which the forward drop is  $V_{D0} = 0.7$  V.

- i. Show the input and output waveforms. [4]  
ii. Draw the VTC curve [2]

## Problem-1

- ① Rectifier converts AC to DC. It operates using a diode's property of allowing current flow in one direction.



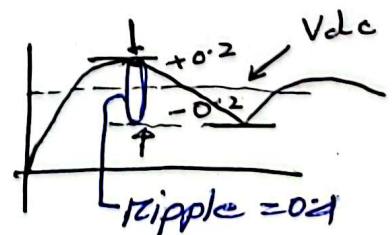
$$\begin{aligned} \text{③ } DC \text{ Avg} &= 2 \frac{V_{op}}{\pi} \\ &= 2 \times 0.318 \times 3.6 \\ &= 2.2896 \end{aligned}$$

$$\begin{aligned} V_{op} &= V_{inp} - 2V_{D0} \\ &= 5 - 2 \times 0.7 \\ &= 3.6 \end{aligned}$$

④  $V_{dc} \pm 0.2 = V_{out}$

$$V_{R(p-p)} = 0.4$$

$$V_{R(p-p)} = \frac{V_{op}}{2Rf_{in}C}$$



$$\Rightarrow 0.4 = \frac{3.6}{2 \times 60 \times C \times 10^3}$$

$$\Rightarrow C = \frac{3.6}{2 \times 60 \times 0.4 \times 10^3}$$

$$\therefore C = 75 \text{ } \overset{\text{MF}}{\cancel{\mu\text{F}}} \text{ (} \overset{\text{MF}}{\cancel{\mu\text{F}}}\text{)}$$

$$\textcircled{e} \quad V_{dc \text{ avg}} = V_{op} - \frac{V_{r(p-p)}}{2}$$

$$= 3.6 - \frac{0.4}{2}$$

$$= 3.4$$

$$3.4 > 2.2896 \text{ (from C)}$$

much better than before.

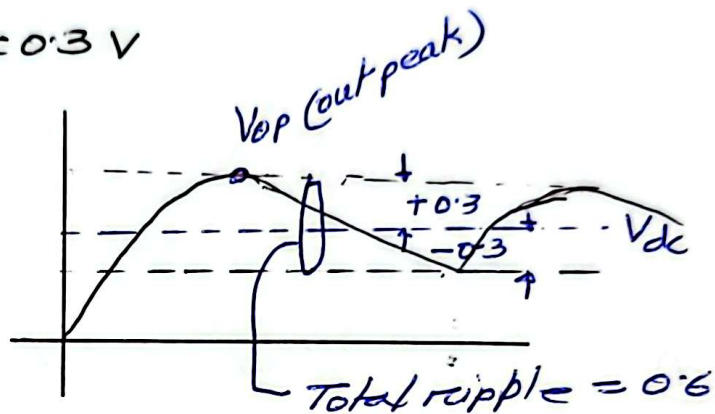
## Problem-2

①  $V_{in} = 10 \sin(2\pi 55t)$

$$V_{dc} = \frac{V_{op}}{\pi} = \frac{V_{inp} - 0.7}{\pi} = \frac{10 - 0.7}{\pi} = \underline{2.7574}$$

②  $V_{out} = V_{dc} \pm 0.3 \text{ V}$

again,



↑ Figure is for understanding

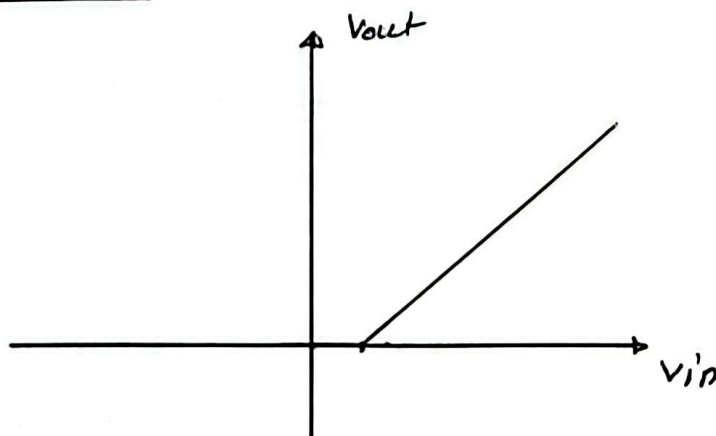
$$V_{r(p-p)} = 0.6$$

$$0.6 = \frac{V_{op}}{R f_{in} C} \Rightarrow 0.6 = \frac{9.3}{2.5 \times 10^3 \times 55 \times C}$$

$$C = \frac{9.3}{2.5 \times 10^3 \times 55 \times 0.6} = 11.27 \times 10^{-6} \text{ F}$$

$$= 11.27 \mu\text{F}$$

③ VTC of HW:



### Problem-3

Ⓐ Input frequency = 200 Hz  $[200\pi = 2\pi f]$   
Output frequency = 400 Hz  $[FW - Out = 2in]$

Ⓑ ~~see~~ similar to problem 1.

Ⓒ  $C = 100\mu F$   
 $R = 5k\Omega$   
 $f_{in} = 200\text{ Hz}$

$$V_{op} = V_{inp} - 2V_{D0} = 7 - 2 \times 0.3 = 6.4$$

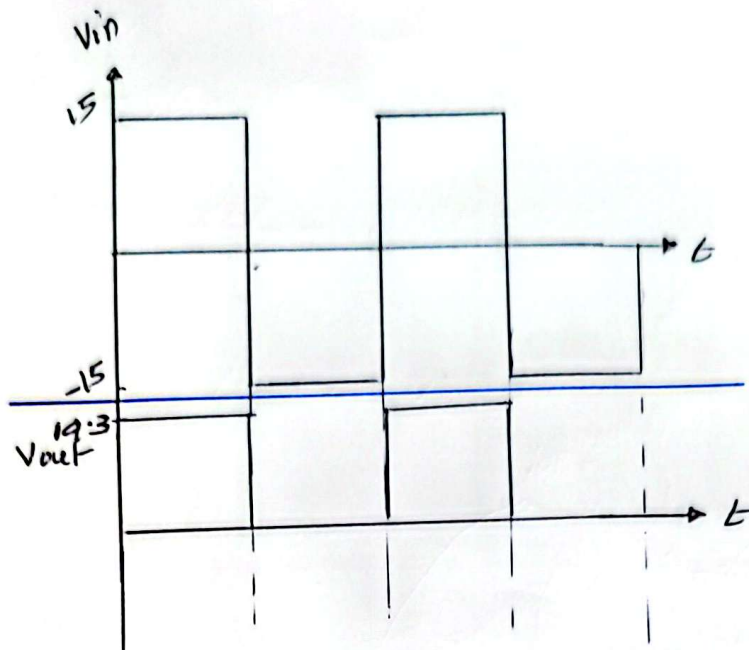
$$V_{R(p-p)} = \frac{V_{op}}{2RfC} = \frac{6.4}{2 \times 5 \times 10^3 \times 200 \times 100 \times 10^{-6}}$$
$$= 0.664 \text{ V}$$
$$= \underline{\underline{64 \text{ mV}}}$$

Ⓓ Better filtering can be provided by reducing ripple voltage. If input diodes used & load is unchanged this can be achieved using higher value capacitor.

Ⓔ Ripple frequency is same as output frequency.

### Problem - 4

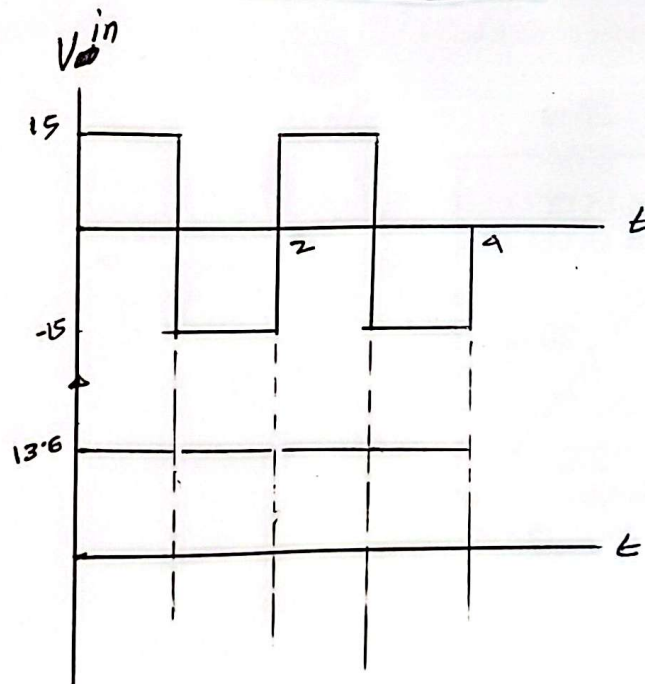
⑦



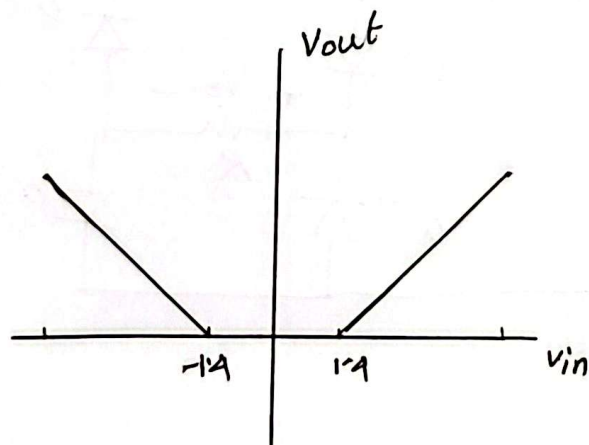
⑪ same as before.

### Problem - 5

⑦



⑪





CSE251  
Electronic Devices & Circuits  
Brac University  
Spring 2025

Set:

A

Quiz: 3  
Marks: 20  
Time: 25 Minutes  
Date: 15<sup>th</sup> April 2025

|               |  |         |             |  |
|---------------|--|---------|-------------|--|
| Student Name: |  | Sec: 02 | Student ID: |  |
|---------------|--|---------|-------------|--|

### Question 1 (CO3) [10]

You have a mobile phone that needs 12V to charge. The phone can handle  $\pm 5\%$  voltage fluctuations while charging. Now you have access to diodes that have  $V_{DO} = 0.75V$  & an AC source that can generate signals with **up to 20V** amplitude, with **60Hz** frequency. Design a rectifier with or without capacitor so that you can charge your phone.

Show the circuit diagram with proper labeling & necessary calculations.

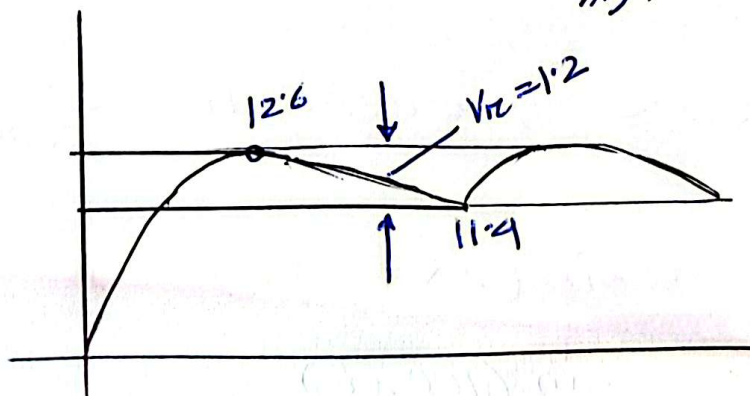
|                                                       |                         |
|-------------------------------------------------------|-------------------------|
| What the equation of the input signal from AC supply? | $19.1 \sin(120\pi t)$   |
| What is the value of the capacitor?                   | $0.0875 \times 10^{-4}$ |
| What is the value of peak output voltage?             | 12.6                    |
| What is the ripple voltage?                           | 1.2                     |

①  $V_{dcavg}$  has to be 12V.

now,  $\pm 5\%$  fluctuations means output can be  $12 + 5\%$   
 $12 - 5\%$

5% of 12 is  $12 \times 0.05 = 0.6V$

so, output can vary from  $\frac{12.6}{\text{high}} \sim \frac{11.4}{\text{low}}$



AC source voltage is 60Hz.

$$V_{RC(P-P)} = \frac{V_{op}}{2f_{in} R C}$$

$$V_{op} = 12.6 \text{ (Ans. (iii))}$$

$$f_{in} = 60$$

$$R = X \text{ [assume]}$$

$$\Rightarrow 1.2 = \frac{12.6}{2 \times 60 \times X \times C}$$

$$V_{RC} = 0.12 \text{ (Ans. (iv))}$$

$$\Rightarrow C = \frac{12.6}{120 \times 1.2 \times X}$$

$$\therefore C = 0.0875 X^{-1} \text{ (Ans. (v))}$$

$$\begin{aligned} V_{in_p} &= V_{op} + 2V_{o0} = 12.6 + 2 \times 0.75 \\ &= 14.1 \end{aligned}$$

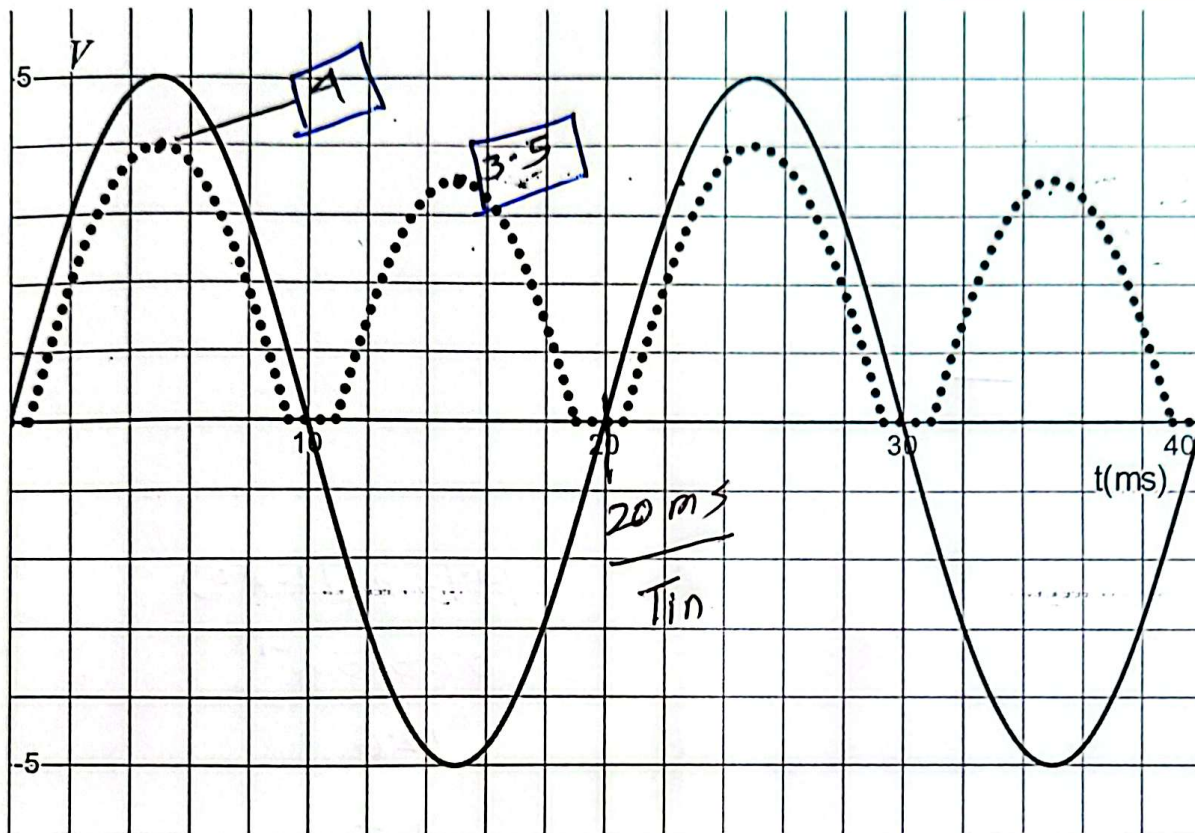
$$V_{in} = 14.1 \sin(2 \times \pi \times 60 \times t)$$

$$V_{in} = 14.1 \sin(120\pi t) \text{ (Ans. (i))}$$

**Question 2 (CO2) [10]**

Dashed Line – Output Voltage

Solid Line- Input Voltage



From the graph above draw the rectifier circuit. Find the input wave's equation. The voltage drop of the diodes used. Finally draw the circuit.

2 different Peaks. So the diodes have different  $V_{D0}$ .

For Pos half cycle,  $5 - 4 = 1$ ,  $V_{D0} = \frac{1}{2} = 0.5$

Neg " " ,  $5 - 3.5 = 1.5$ ,  $V_{D0} = \frac{1.5}{2} = 0.75$

input peak = 5 frequency in =  $\frac{1}{20\text{ms}} = 50\text{Hz}$

$$V_{in} = 5 \sin(2 \times 50 \pi t)$$

$$= 5 \sin(100 \pi t)$$

